

Preface

UX MEANS USER EXPERIENCE

Welcome to *The UX Book*, Third Edition. We'd like to begin by making sure everyone knows that UX is short for "user experience." In simple terms, user experience is the phenomenon felt by users before, during, and after usage—usually a combination of usability, usefulness, emotional impact, and meaningfulness.

GOALS FOR THIS BOOK

Imbue an understanding of what a good user experience is and how to achieve it.

Our main goal for this book is simple: to help readers learn to recognize, understand, and design for a quality user experience. Sometimes, a quality user experience is like an electric light: When it works, nobody notices. Sometimes, a user experience is really good and is noticed and even appreciated, and it leaves pleasant memories. Or sometimes, the effects of a bad user experience persist in the user's mind. So, early in the book, we talk about what constitutes a positive user experience.

Emphasize design. What is less obvious than understanding a quality user experience is how to design for it. In this edition, we continue to stress design—a kind of design that highlights the designer's creative skills and insights and embodies a synthesis of technology with aesthetics and meaningfulness to the user. In Part 3, we show you a variety of design methods to help you get just the right approach for your project.

Give a how-to approach. We have designed much of the book as a how-to handbook and field guide, as a textbook for students aspiring to be UX professionals and for professionals aspiring to be better. The approach is practical, not formal or theoretical. Some references are made to the related science, but they are usually to provide context for the practice and are not necessarily elaborated.

Provide a good reader experience. Beyond our main goal of helping readers learn about UX and UX design, other goals for your experience include ensuring that:

- You get excited about UX design.
- The material is easy to learn.
- The material is easy to apply.
- The material is useful to students and professionals.
- For you, the reader, the experience is satisfying and at least a little bit fun.

Provide broad coverage. Our goals for scope of coverage include:

- Depth of understanding—detailed information about different aspects of the UX process (like having an expert accompanying the reader)
- Breadth of understanding—as comprehensive as space permits
- Broad range of application—the process and the design infrastructure and vocabulary, including guidelines, are not just for graphical user interfaces and the web, but for all kinds of interaction styles and devices, including ATMs, refrigerators, road signs, ubiquitous computing, embedded computing, and everyday things.

USABILITY IS STILL IMPORTANT, BUT USER EXPERIENCE GOES BEYOND

Usability, a key component of a quality user experience, is still an essential part of what we have to say about UX design. It is about designing technology as an extension of human capabilities to accomplish something and to be as transparent as possible in the process.

We'll see that user experience transcends traditional usability attributes, entailing social and cultural values, and emotional impact—how the interaction experience includes joy of use, fun, aesthetics, and meaningfulness in the lives of users.

A PRACTICAL APPROACH

This book takes a practical, applied, hands-on approach based on the application of established and emerging best practices, principles, and proven methods to ensure a quality user experience. Our approach is about practice, drawing on the creative concepts of design exploration and visioning to make designs that appeal to the emotions of users, while moving toward processes

that are lightweight, rapid, and agile—to make things as good as resources permit, and not waste time and other resources in the process.

PRACTICAL UX METHODS

In the second edition, we described some rigorous methods as well as many rapid methods for agile use. UX designers who need rigorous methods to develop large-scale systems in complex domains will still find what they need in this book. Following this trend, the third edition reflects the fact that agile methods now dominate most UX practice. We have adopted a practical stance to balance the need for rigor with practical and economical processes and methods that will suit a very large percentage of all projects.

FROM AN ENGINEERING ORIENTATION TO A DESIGN ORIENTATION

For a long time, the focus in human–computer interaction (HCI) practice was on engineering, taking its cues from usability engineering and human factors engineering. In this edition, we continue to lean away from a pure engineering focus to more of a design and synthesis focus.

AUDIENCES

Anyone involved in, or wishing to learn more about, creating products and systems that engender a quality user experience will benefit from this book. One important audience is students and the instructors who teach them. Another important intended audience includes UX practitioners: UX professionals or others who take on the role of a UX professional in project environments. The perspective of professionals is very similar to that of students; both have the goal of learning, only in slightly different settings and perhaps with different motivations and expectations.

Our readership comprises all kinds of UX professionals, including UX designers, content strategists, information architects, graphic designers, web designers, usability engineers, mobile device app designers, usability analysts, human factors engineers, cognitive psychologists, cosmic psychics, trainers, technical writers, documentation specialists, marketing personnel, and project managers. Readers in any of these areas will find the hands-on approach of this book valuable and can focus mainly on the how-to parts.

Software people who work with UX professionals—including software engineers, programmers, systems analysts, software quality-assurance specialists—can also benefit from this book. Software engineers who find themselves called upon to do some UX design will find this book easy to read and apply because UX design lifecycle concepts are similar to those in software engineering.

WHAT'S CHANGED SINCE THE SECOND EDITION?

Readability

Much of the effort we invested in making the third edition was aimed at better readability. Toward this end each chapter has undergone a large amount of editing—an average of over 12 versions per chapter, with one chapter topping out at 23 versions! This extensive editing for readability was a quest for:

- Simple, straightforward sentence structure
- Precision of word choices
- Clarity of thought
- Coherence of content within paragraphs

Good readability also demands tight writing. We tightened our writing by turning verbosity, meandering, and repetition into concise presentation of content. In addition, two factors allowed us to abbreviate some of the basic material and eliminate substantial amounts of descriptive (and prescriptive) text:

- UX practice in the field has become more streamlined.
- Common knowledge of the basics has risen with maturation of the field, requiring less explanation of the lowest-level details.

As an additional benefit from tight writing for readability, we achieved reduced volume and weight of the book.

Content

Most of our changes in content were to bring it in line with what practitioners need today. For example, we updated the UX research chapters, bringing them more in line with current agile practice in the field. UX practice in the field has evolved to rely less on detailed, formal processes and is more about choices for UX teams to adapt to different project environments and design situations. Correspondingly, the process chapters now reflect a more mature

process, more closely aligned with what new UX practitioners will need to know.

Here are some examples. We updated the UX research chapters, especially the sections on work, work practice, and work domains, to better prepare readers for the reality of practice. In UX research data analysis, we now get into the organization and synthesis of data from multiple sources earlier and more directly. UX research data modeling is more streamlined and more directly useful in applying what is learned in UX research activities to UX design. A more realistic concept of user stories is one dimension of what is now used by UX practitioners to drive design.

In the design activity, we present a more balanced approach with enhanced treatment of top-down design, as a force to foster innovation. In this context the concept of abstract work activities allows designers to avoid the biases of existing designs and focus on the essence of what a design should accomplish.

Also, we have introduced an additional running example about camping reservation within the US National Park Service, illustrating various components of UX design, especially interaction design.

We have also introduced the use of formal summative evaluation methods that have become mainstream in UX design and evaluation practice since the second edition, including A/B comparison testing and customer satisfaction surveys.

Finally, we have reduced coverage of rigorous methods in evaluation that are no longer popular in the current age of agile processes, and we updated the reporting chapter to make it more actionable for trainee practitioners.

A more relaxed approach to grammar and writing style

For the sake of readability, we continue with some changes in the mechanics of writing that we initiated in the previous edition. We moved toward a more informal, and hopefully more readable, style by relaxing some grammatical rules. We repeat them here for those not familiar with the previous edition.

Good grammar is important. However, some of the old grammatical constraints make the writing and reading awkward. For example, many people now feel that split infinitives are often easier on the ear (to more casually communicate) than the more formal alternative.

Most of those rules were never real, anyway. In case you are a stickler about these rules, they may not be what you think. It is not that the language has evolved through modern usage to make the breaking of these rules the new norm (although that has certainly happened). It is more that these so-called rules never were real (O’Conner & Kellerman, 2013). Beyond that (for the

real sticklers), there is nothing to split in an infinitive, because the word *to* is technically not part of the infinitive. Famous authors such as Shakespeare, Donne, and Wordsworth have been splitting infinitives since the 1200s. So, our hope is we will not be faulted for easing up on some grammatical rules thought to be ironclad but which are really optional. See also Pinker's (2014) book, *The Sense of Style*.

Ways we are relaxing the grammar. We may occasionally allow a sentence to begin with a conjunction. And that's perfectly good when needed. We sometimes find a preposition a good thing to end a sentence with. We're also now in the habit of using contractions. By some standards, they are verboten in published writing, but please agree that informal speech flows nicely with contractions, and we see no reason you can't experience that same pleasant cadence in your reading.

Another example of something you're not supposed to do is have a one-sentence paragraph, but sometimes it's a nice way to highlight a point.

In addition, we use personal pronouns, a practice not usually admired in formal writing but, well, we're trying to be less formal. So, instead of saying, "the reader will notice..." we'll use the less formal and friendlier, "you will notice..." Similarly, we'll use "we" or "us" to refer to ourselves, the authors. Sometimes, we'll say "I" or "me," and you may have to guess which of us that means.

Another area where we will be relaxed is anthropomorphic wording, which might cause some grammarians to roll over in their ivory towers, but sometimes it's just easier, and readers will always know what we mean. For example, we might say, "Figure X shows such and such," when, of course, we mean, "in Figure X, we show such and such." I'll bet that if we hadn't mentioned that one here, most readers wouldn't notice.

Gender-specific pronouns. We have observed that literary acrobatics to avoid the use of gender-specific pronouns can lead to considerable stiffness in writing. We are saddled with a language that has built-in gender traps that are difficult and awkward to avoid. Sometimes, you can just use the plural—for example, instead of "the user must type Ctrl-Shift-Alt-Delete often, and he will find that difficult," you can say, "users must type Ctrl-Shift-Alt-Delete often, and they will find that difficult." Other approaches to de-gendering can result in graceless, lashed-together combinations, such as he/she or s/he or the ungainly and mysterious "one," as in "When one takes on the role of UX professional, then one must be very brave." "We have tried to be relaxed about he, she, they, and them."

Relaxing about references. Most of the references cited will be in a standard academic style. But sometimes, a less formal source is sufficient and practical.

In particular, occasionally we'll use a reference to Wikipedia for a “definition” of a commonly used term. Yes, sometimes Wikipedia or another website and other similar sources can be flawed, but we will do our best to avoid anything questionable.

Also, this. We shall struggle to never use the word “plethora.”

WHAT WE DON'T COVER

To begin with, this book is not a survey of the HCI field, nor is it a survey about user experience. It is also not about HCI research. Although this book is quite comprehensive, we could not hope to include everything about HCI or UX. We apologize if your favorite topic is excluded, but we had to draw the line somewhere. Further, many of these additional topics are so broad in themselves that they could (and most do) fill a book of their own.

Among the specific topics not included are:

- Accessibility, special needs, and the Americans with Disabilities Act
- Internationalization and cultural differences
- Standards
- Ergonomic health issues, such as repetitive stress injury
- Specific HCI application areas, such as societal challenges, health care systems, help systems, training, and designing for elders or other special user populations
- Special areas of interaction, such as virtual environments or 3D interaction
- Computer-supported collaborative work
- Social media
- Personal information management
- Sustainability
- Artificial intelligence (AI)

ABOUT THE EXERCISES

After certain topics are covered in this book, we present exercises to let you try your hand at the concepts. The exercises are an integral part of the learning from this book. In its use of hands-on exercises, the book is somewhat like a workbook.

Some of the exercises, especially the early ones, are based on a hypothetical system called the Ticket Kiosk System (TKS), which is used as a running UX design example to illustrate the material throughout the process chapters. We use the TKS example system to show how you can use the activities described

in the book to build your own designs. In other examples, especially the later ones, we ask you to find a project of your own, hypothetical or real, on which you can base your exercises.

After each main topic, you get to apply the new material immediately, learning the practical techniques by active engagement in their application. The book is organized and written to support active learning (learning by doing), and should be used that way.

The exercises require medium-level engagement, somewhere in between the in-text examples and full project assignments.

Take them in order. Each process chapter builds on the previous ones and adds a new piece to the overall puzzle. Each exercise builds on what you learned and accomplished in the previous stages—just as in a real-world project.

Do the exercises as a team if you can. Good UX design is almost always a collaborative effort. Working through the exercises with at least one other interested person will enhance your understanding and learning of the materials greatly. In fact, many of the exercises are written for small teams (e.g., three to five people) because they involve multiple roles.

The teamwork will help you understand the kinds of communication, interaction, and negotiation that take place in creating and refining a UX design. If you can season the experience by including a software developer with responsibility for software architecture and implementation (at least for a working prototype), then many important communication needs will become apparent.

Students should do the exercises as a team in the classroom. If you are a student taking a course, the best way to do the exercises is as team-based, in-class exercises. The exercises are adapted easily for classroom use as an ongoing, semester-long interactive set of in-class activities to understand needs, design solutions, prototype candidate designs, and evaluate the user experience. The instructor can observe and comment on your progress, and you can share your “lessons learned” with other teams.

UX professionals should get buy-in to do the exercises at work. If you are a UX professional, or you aspire to learn on the job to be a UX professional, trying to learn this material in the context of your regular work, the best way is an intensive short course with team exercises and projects. We used to teach just such a short course.

Alternatively, if you have a small UX team in your work group (perhaps a team that expects to work together on a real project), and your work environment allows, set aside some time (say, 2 hours every Friday afternoon)

for the team exercises. To justify the extra overhead to pull this off, you may have to convince your project manager of the value added.

Individuals can still do the exercises. Do not let the lack of a team stop you from doing the exercises. Try to find at least one other person with whom you can work, or, if necessary, get what you can from the exercises on your own. Although it would be easy to let yourself skip the exercises, we urge you to do as much on each of them as your time permits.

INSTRUCTOR'S GUIDE

Information on accessing the Instructor's Guide is available on the book website: <https://www.theuxbook.net/>. The Instructor's Guide contains teaching materials tailored specifically to this book, including a sample course syllabus, sample course calendar, sample sets of lecture slides for many of the chapters, and much more.

TEAM PROJECTS

Students. In addition to the small-scale series of in-class activities in conjunction with the book exercises, we offer fully detailed and more involved team projects. We consider this semester-long sequence of team projects an essential part of learning by doing in a course based on this book. These team projects have always been the most demanding, but also the most rewarding, learning activity in the course.

For this semester-long team project, we recommend using real clients from the community—a local company, store, or organization that needs some kind of interactive software application designed. The client stands to get some free consulting and even (sometimes) a system prototype in exchange for serving as the project client. A sample set of team project assignments is available in the Instructor's Guide, available to instructors from the publisher.

UX professionals. As a way of getting started in applying this material to your real work environment, you and your coworkers can select a low-risk but real project. Your team may already be familiar and even experienced with some of the activities we describe and may even already be doing some of them in your development environment. By making them part of a more complete and informed development lifecycle, you can integrate what you know with new concepts presented in the book.

ABOUT THE AUTHORS

Rex Hartson is a pioneer researcher, teacher, practitioner, and consultant in HCI and UX. He is the founding faculty member of HCI (in 1979) in the Department of Computer Science at Virginia Tech, from where he is now retired. Hartson has been principal investigator (PI) or co-PI at Virginia Tech on a large number of research grants and has published many journal articles, conference papers, and book chapters. He has presented many tutorials, invited lectures, workshops, seminars, and international talks. His HCI practice is grounded in decades of consulting and UX design training for dozens of clients in business, industry, government, and the military.

Pardha S. Pyla is the Head of UX of Core Data at Google. He is an award-winning designer and strategist with deep expertise in envisioning and delivering industry-leading products. He is a seasoned leader with a record of establishing exemplary user experience and product development practices. He is a pioneering researcher in the area of coordinating software engineering and UX lifecycle processes and the author of several peer-reviewed research publications in human–computer interaction and software engineering. He has received numerous awards in recognition of his work in design, research, teaching, leadership, and service.